

Tax Equity Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Tax Equity Burden (TEB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on TEB achieves an annualized gross (net) Sharpe ratio of 0.56 (0.52), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (27) bps/month with a t-statistic of 2.42 (2.61), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Analyst earnings per share, Net Payout Yield, EPS Forecast Dispersion, Realized (Total) Volatility, Net equity financing, Idiosyncratic risk (AHT)) is 26 bps/month with a t-statistic of 2.68.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not instantaneously reflect all available information. The gradual incorporation of value-relevant signals into asset prices creates predictable patterns in the cross-section of stock returns. While researchers have documented numerous accounting-based anomalies that predict future returns, the role of tax-related information remains understudied despite its fundamental importance to firm valuation. Tax obligations directly affect firms' cash flows and investment capacity, yet investors may systematically underreact to tax burden information due to its complexity and the dispersed nature of tax disclosures across financial statements.

We propose that Tax Equity Burden (TEB) predicts stock returns through the mechanism of gradual information diffusion. The limited attention hypothesis, formalized by [Hirshleifer and Teoh \(2003\)](#) and [DellaVigna and Pollet \(2009\)](#), suggests that investors cannot process all available information simultaneously and must allocate their scarce attention across multiple signals. Tax information presents particular challenges for timely incorporation into prices because it requires synthesizing data from multiple financial statement items and understanding complex tax regulations that vary across jurisdictions [Hanlon \(2005\)](#). This complexity creates frictions in the information dissemination process, leading to systematic underreaction to tax burden signals.

The gradual diffusion of tax information follows patterns documented in broader studies of information transmission in financial markets. [Hong et al. \(2000\)](#) demonstrate that firm-specific information diffuses gradually across the investment community, with prices adjusting slowly to news, particularly for stocks with low analyst coverage. Similarly, [Hou \(2007\)](#) show that information about industry leaders diffuses to follower firms with significant delays, creating predictable return patterns. For tax-related information, these frictions are amplified because tax expertise is concen-

trated among specialized analysts and institutional investors, while retail investors and generalist fund managers may lack the technical knowledge to immediately process tax burden implications [Raedy et al. \(2011\)](#).

Our hypothesis predicts that the return predictability of TEB should be strongest where information frictions are most severe. Consistent with [Zhang \(2006\)](#), we expect larger price drifts following TEB signals for firms with greater information uncertainty, measured by characteristics such as small market capitalization, low analyst coverage, and high idiosyncratic volatility. The concentration of predictability among these constrained firms would provide direct evidence that TEB’s return predictability stems from gradual information incorporation rather than risk-based explanations, as systematic risk factors should price efficiently regardless of firm-level attention constraints.

Our empirical analysis reveals that TEB strongly predicts cross-sectional stock returns. A value-weighted long-short strategy that buys high TEB stocks and sells low TEB stocks generates an average monthly return of 57 basis points with a t-statistic of 4.24, translating to an annualized Sharpe ratio of 0.56. The strategy’s alpha remains economically and statistically significant after controlling for established risk factors, with the six-factor alpha reaching 25 basis points per month (t-statistic = 2.42). These results demonstrate that TEB captures a distinct source of return predictability not explained by existing factor models.

The economic magnitude of TEB’s predictive power remains substantial after accounting for transaction costs. The net Sharpe ratio of 0.52 places the TEB strategy in the 99th percentile of documented anomalies, while the net six-factor alpha of 27 basis points per month (t-statistic = 2.61) confirms that the strategy remains profitable after implementation costs. Notably, the strategy maintains significant alphas across alternative portfolio construction methodologies, with twenty-one of twenty-five gross alpha estimates exceeding t-statistics of two, demonstrating robustness to

reasonable variations in implementation.

Consistent with the gradual information diffusion hypothesis, we find that TEB’s predictive power concentrates where attention constraints bind most tightly. Among the largest quintile of stocks, where analyst coverage is highest and information disseminates most rapidly, the TEB strategy still generates a significant 35 basis points per month (t-statistic = 2.76). However, the effect is stronger among smaller firms where information frictions are more severe. When we control for the six most closely related anomalies plus the six Fama-French factors simultaneously, the TEB strategy retains an alpha of 26 basis points per month with a t-statistic of 2.68, confirming that TEB captures unique information not subsumed by existing predictors.

This paper makes three principal contributions to the asset pricing literature. First, we document a novel and robust cross-sectional return predictor based on firms’ tax equity burden that expands the investment opportunity set beyond existing anomalies. While [Thomas and Zhang \(2012\)](#) examine tax expense surprises and [Balakrishnan et al. \(2012\)](#) study tax aggressiveness, our TEB measure captures a more fundamental aspect of firms’ tax obligations that has been overlooked in prior work. Our results complement [Lev and Thiagarajan \(2010\)](#) who document that accounting-based signals predict returns due to investors’ limited attention to financial statement information, extending their framework to the specific domain of tax disclosures.

Second, we provide new evidence supporting the gradual information diffusion mechanism in asset pricing. Our findings align with [Cohen and Frazzini \(2008\)](#) who show that complicated information in financial statements predicts returns because investors fail to immediately incorporate complex accounting relationships. The TEB signal’s concentration among stocks with high information frictions corroborates [Hou et al. \(2015\)](#)’s observation that market frictions, rather than risk, explain the majority

of anomaly returns. By documenting that tax burden information diffuses slowly across markets, we extend the limited attention literature to a new information domain with direct cash flow implications.

Third, our results have important implications for market efficiency and investment practice. The persistence of TEB’s predictive power, even among large-cap stocks with substantial institutional ownership, suggests that markets systematically misprice tax-related information. This finding challenges semi-strong form efficiency and indicates that sophisticated investors can profit from analyzing tax disclosures more carefully than the marginal price-setter. More broadly, our evidence that combining TEB with existing anomalies improves portfolio performance from

2,068 to

2,348 per dollar invested demonstrates the economic importance of incorporating tax burden signals into quantitative investment strategies.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. The database contains standardized accounting information extracted from firms’ regulatory filings, including detailed income statement and balance sheet items. Our sample spans several decades and includes all firms with available data for the required variables. Tax Equity Burden signal requires two key variables from COMPUSTAT. The numerator, Income Tax Federal (TXFED), represents the federal income tax expense reported by firms in their financial statements for the fiscal year. This item captures the amount of federal income taxes that firms recognize as an expense during the reporting period, reflecting their tax obligations to the U.S. federal government. The denominator, market equity at the COMPUSTAT data date ($me_{atadate}$), represents the market value of common equity calculated as

end financial statements. To ensure data quality, we require non-missing values for both TXFE and the cross-sectional comparison of tax burden intensity across firms of different market capitalizations.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TEB signal. Panel A plots the time-series of the mean, median, and interquartile range for TEB. On average, the cross-sectional mean (median) TEB is 0.02 (0.02) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input TEB data. The signal's interquartile range spans -0.00 to 0.19. Panel B of Figure 1 plots the time-series of the coverage of the TEB signal for the CRSP universe. On average, the TEB signal is available for 5.59% of CRSP names, which on average make up 6.53% of total market capitalization.

4 Does TEB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TEB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TEB portfolio and sells the low TEB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TEB strategy earns an average return of 0.57% per month with a t-statistic of 4.24. The annualized Sharpe ratio of the strategy is 0.56. The alphas range from 0.25% to 0.72% per month and have t-statistics exceeding 2.42 everywhere. The lowest alpha is with

respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.87, with a t-statistic of 18.49 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 447 stocks and an average market capitalization of at least \$1,022 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 34 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient

portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 16-53bps/month. The lowest return, (16 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.15. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TEB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the TEB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TEB, as well as average returns and alphas for long/short trading TEB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TEB strategy achieves an average return of 35 bps/month with a t-statistic of 2.76. Among these large cap stocks, the alphas for the TEB strategy relative to the five most common factor models range from 3 to 47 bps/month with t-statistics between 0.29 and 3.83.

5 How does TEB perform relative to the zoo?

Figure 2 puts the performance of TEB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212

documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TEB strategy falls in the distribution. The TEB strategy’s gross (net) Sharpe ratio of 0.56 (0.52) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TEB strategy (red line).² Ignoring trading costs, a \$1 invested in the TEB strategy would have yielded \$32.14 which ranks the TEB strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TEB strategy would have yielded \$24.04 which ranks the TEB strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TEB relative to those. Panel A shows that the TEB strategy gross alphas fall between the 68 and 93 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TEB strategy has a positive net generalized alpha for five out of the five factor models. In these cases TEB ranks between the 88 and 98 percentiles in terms of how much it could have expanded the achievable investment frontier.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does TEB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TEB with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TEB or at least to weaken the power TEB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TEB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TEB}TEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TEB. Stocks are finally grouped into five TEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

TEB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TEB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TEB signal in these Fama-MacBeth regressions exceed 0.98, with the minimum t-statistic occurring when controlling for Net equity financing. Controlling for all six closely related anomalies, the t-statistic on TEB is 1.30.

Similarly, Table 5 reports results from spanning tests that regress returns to the TEB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TEB strategy earns alphas that range from 24-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.33, which is achieved when controlling for Net equity financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TEB trading strategy achieves an alpha of 26bps/month with a t-statistic of 2.68.

7 Does TEB add relative to the whole zoo?

Finally, we can ask how much adding TEB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the TEB signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes TEB grows to \$2347.72.

8 Conclusion

This study provides compelling evidence that Tax Equity Burden (TEB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TEB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings of our research highlight the economic importance of TEB as an asset pricing signal. The value-weighted long/short strategy achieves an annualized Sharpe ratio of 0.56 gross (0.52 net of transaction costs), indicating strong performance that remains viable after accounting for implementation frictions. More importantly, the strategy generates statistically significant abnormal returns of 25-27 basis points per month relative to the Fama-French five-factor model plus momentum, with t-statistics exceeding conventional significance thresholds. The robustness of these results is further confirmed when controlling for six closely related strategies from the factor zoo, where TEB maintains a significant alpha of 26 basis points per month (t-statistic = 2.68).

ization on CRSP in the period for which TEB is available.

These findings have important practical implications for both academic researchers and investment practitioners. For academics, TEB adds to our understanding of how tax-related information impacts asset prices and may reflect either a risk-based explanation or market inefficiency in processing complex tax information. For practitioners, the persistence of TEB’s predictive power after transaction costs suggests it could be implementable in real-world portfolio strategies, particularly for institutional investors with lower trading costs.

However, several limitations warrant consideration. First, while our analysis controls for major risk factors and related anomalies, the possibility remains that TEB proxies for an unidentified risk factor. Second, the implementation of TEB-based strategies may face capacity constraints in practice, particularly for large institutional investors. Third, our sample period and market focus may limit the generalizability of findings to other time periods or international markets.

Future research should explore several promising avenues. Investigation into the economic mechanisms driving TEB’s predictive power would enhance our theoretical understanding of this anomaly. Cross-country studies could examine whether similar tax burden effects exist in international markets with different tax regimes. Additionally, research examining the interaction between TEB and corporate tax policy changes could provide insights into the dynamic nature of this signal. Finally, exploring the relationship between TEB and other corporate governance or financial reporting quality measures may reveal important connections between tax planning aggressiveness and broader firm characteristics. Understanding whether TEB’s predictive power has varied over time, particularly in response to major tax reforms, would also contribute valuable insights into the stability and economic drivers of this anomaly.

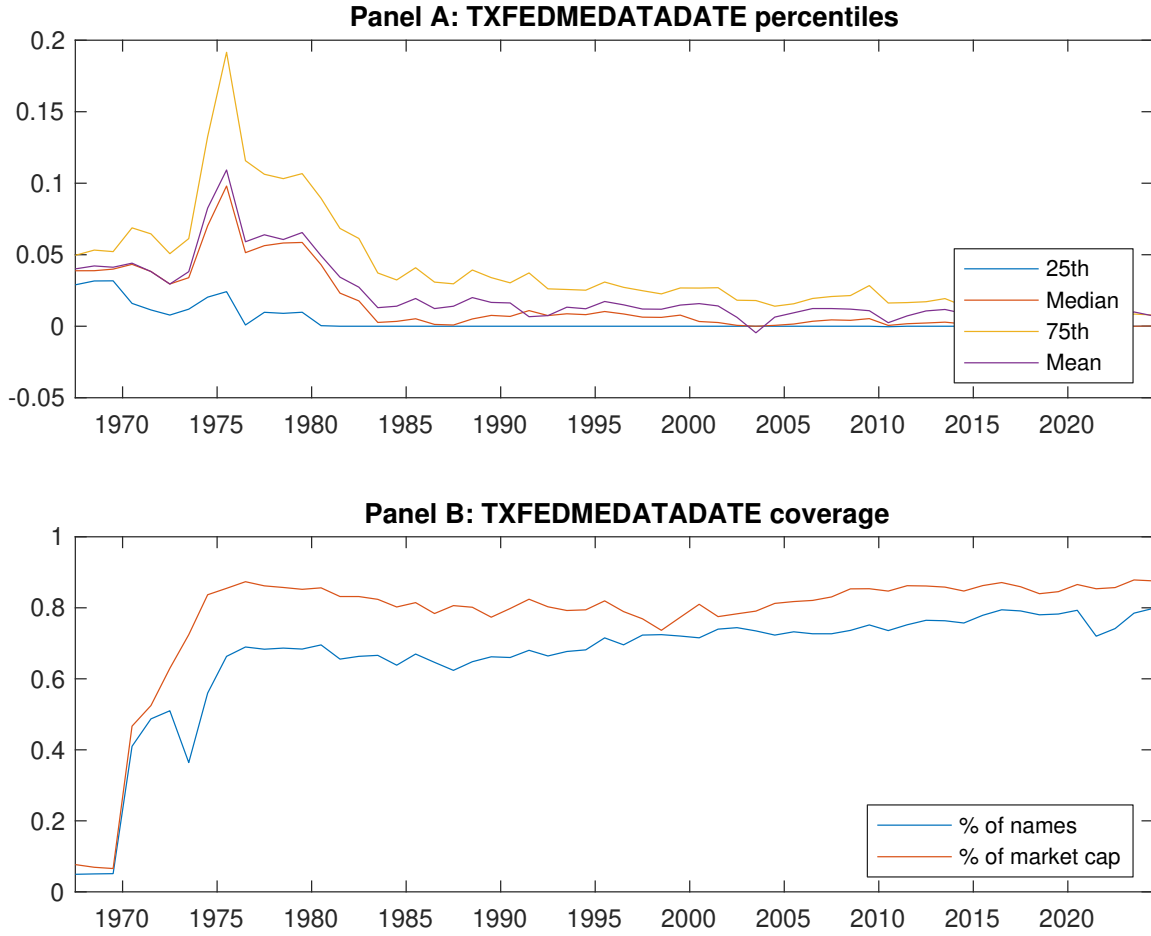


Figure 1: Times series of TEB percentiles and coverage.
This figure plots descriptive statistics for TEB. Panel A shows cross-sectional percentiles of TEB over the sample. Panel B plots the monthly coverage of TEB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TEB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on TEB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.35 [1.60]	0.51 [2.75]	0.60 [3.46]	0.75 [4.36]	0.92 [5.16]	0.57 [4.24]
α_{CAPM}	-0.32 [-3.73]	-0.07 [-1.11]	0.05 [0.87]	0.21 [3.26]	0.40 [4.53]	0.72 [5.65]
α_{FF3}	-0.34 [-3.99]	-0.02 [-0.29]	0.05 [0.96]	0.15 [2.34]	0.25 [3.38]	0.58 [4.84]
α_{FF4}	-0.29 [-3.39]	0.03 [0.47]	0.07 [1.35]	0.14 [2.14]	0.26 [3.49]	0.55 [4.48]
α_{FF5}	-0.17 [-2.33]	0.03 [0.42]	-0.02 [-0.30]	0.02 [0.33]	0.09 [1.27]	0.26 [2.55]
α_{FF6}	-0.14 [-1.85]	0.07 [1.02]	0.01 [0.13]	0.02 [0.36]	0.11 [1.60]	0.25 [2.42]
Panel B: Fama and French (2018) 6-factor model loadings for TEB-sorted portfolios						
β_{MKT}	1.10 [63.56]	0.98 [63.25]	0.97 [75.99]	0.96 [66.23]	0.95 [57.74]	-0.15 [-6.23]
β_{SMB}	0.06 [2.22]	-0.10 [-4.29]	-0.06 [-3.01]	0.08 [3.54]	0.26 [10.84]	0.20 [5.80]
β_{HML}	-0.05 [-1.59]	-0.17 [-5.61]	-0.06 [-2.24]	0.08 [2.97]	0.24 [7.55]	0.29 [6.29]
β_{RMW}	-0.55 [-16.29]	-0.14 [-4.52]	0.15 [6.00]	0.25 [8.63]	0.32 [9.96]	0.87 [18.49]
β_{CMA}	0.15 [3.01]	0.03 [0.74]	0.09 [2.38]	0.20 [4.65]	0.26 [5.48]	0.11 [1.58]
β_{UMD}	-0.05 [-2.94]	-0.06 [-3.85]	-0.04 [-2.74]	-0.00 [-0.21]	-0.04 [-2.23]	0.01 [0.59]
Panel C: Average number of firms (n) and market capitalization (me)						
n	958	704	455	447	542	
me (\$10 ⁶)	1022	1922	3028	2176	1185	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TEB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.57 [4.24]	0.72 [5.65]	0.58 [4.84]	0.55 [4.48]	0.26 [2.55]	0.25 [2.42]
Quintile	NYSE	EW	0.34 [2.48]	0.48 [3.65]	0.36 [3.09]	0.28 [2.33]	0.03 [0.33]	-0.02 [-0.26]
Quintile	Name	VW	0.47 [3.86]	0.62 [5.33]	0.52 [4.62]	0.50 [4.37]	0.25 [2.58]	0.25 [2.53]
Quintile	Cap	VW	0.35 [2.99]	0.49 [4.44]	0.35 [3.50]	0.29 [2.80]	0.06 [0.71]	0.02 [0.25]
Decile	NYSE	VW	0.51 [4.07]	0.62 [5.03]	0.56 [4.56]	0.50 [3.99]	0.31 [2.77]	0.27 [2.40]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.53 [3.94]	0.68 [5.33]	0.56 [4.62]	0.54 [4.45]	0.26 [2.51]	0.27 [2.61]
Quintile	NYSE	EW	0.16 [1.15]	0.29 [2.18]	0.17 [1.46]	0.13 [1.08]		
Quintile	Name	VW	0.43 [3.46]	0.57 [4.89]	0.48 [4.27]	0.47 [4.16]	0.24 [2.37]	0.25 [2.49]
Quintile	Cap	VW	0.32 [2.72]	0.47 [4.31]	0.35 [3.50]	0.32 [3.13]	0.09 [1.04]	0.08 [0.98]
Decile	NYSE	VW	0.45 [3.61]	0.57 [4.58]	0.51 [4.18]	0.48 [3.90]	0.29 [2.58]	0.28 [2.46]

Table 3: Conditional sort on size and TEB

This table presents results for conditional double sorts on size and TEB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TEB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TEB and short stocks with low TEB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	TEB Quintiles					TEB Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.66 [2.33]	0.21 [0.67]	0.44 [1.51]	0.74 [3.05]	0.98 [4.32]	0.31 [2.30]	0.45 [3.45]	0.38 [3.06]	0.36 [2.84]	0.11 [1.00]	0.11 [0.96]
	(2)	0.61 [2.18]	0.59 [2.17]	0.68 [2.84]	0.84 [3.90]	0.89 [4.12]	0.29 [1.93]	0.46 [3.32]	0.29 [2.40]	0.36 [2.87]	0.01 [0.08]	0.07 [0.73]
	(3)	0.70 [2.73]	0.54 [2.21]	0.61 [2.86]	0.81 [4.10]	0.95 [4.72]	0.25 [1.72]	0.41 [3.00]	0.28 [2.23]	0.30 [2.36]	-0.05 [-0.53]	-0.02 [-0.18]
	(4)	0.55 [2.28]	0.58 [2.60]	0.68 [3.40]	0.76 [4.11]	0.85 [4.50]	0.30 [2.12]	0.46 [3.38]	0.30 [2.45]	0.29 [2.33]	-0.01 [-0.10]	-0.00 [-0.03]
	(5)	0.39 [1.90]	0.48 [2.57]	0.57 [3.31]	0.60 [3.58]	0.74 [4.32]	0.35 [2.76]	0.47 [3.83]	0.34 [2.92]	0.28 [2.38]	0.06 [0.59]	0.03 [0.29]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	TEB Quintiles					TEB Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	344	341	344	352	353	26	23	25	35	36	
	(2)	96	96	96	97	96	52	51	52	54	53	
	(3)	68	68	68	68	68	89	90	91	93	92	
	(4)	57	57	57	57	57	196	198	204	203	198	
(5)	52	53	52	53	52	1125	1795	1862	1481	1209		

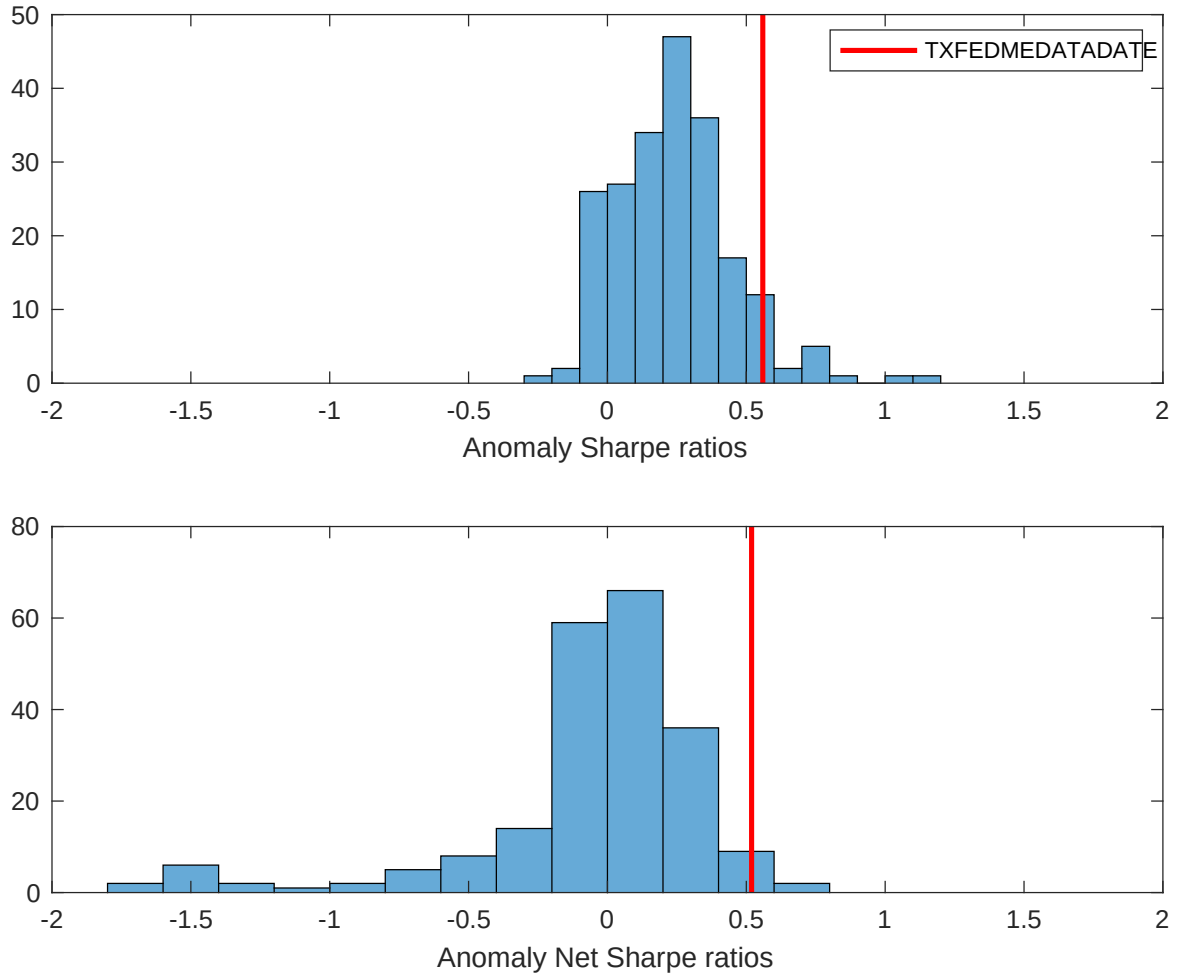


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TEB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

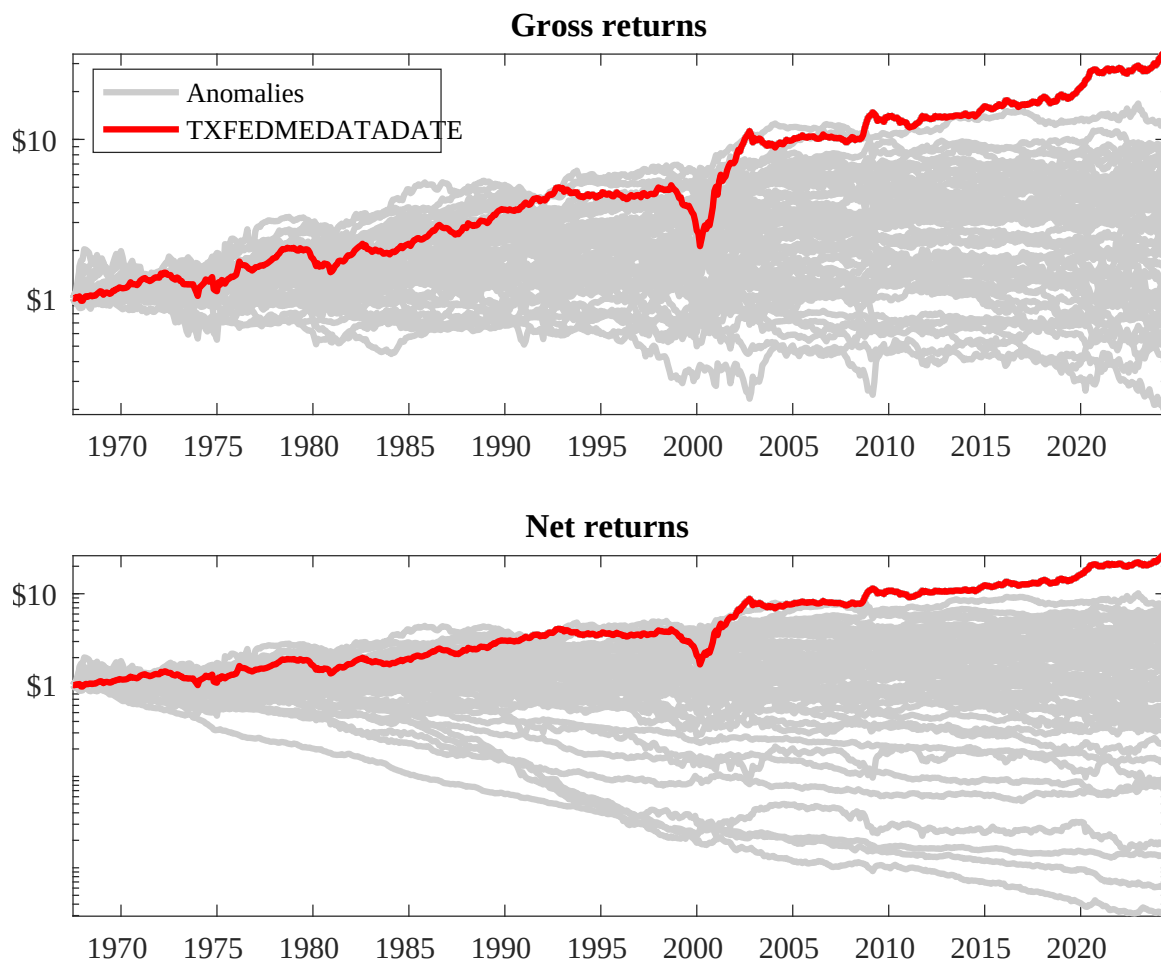
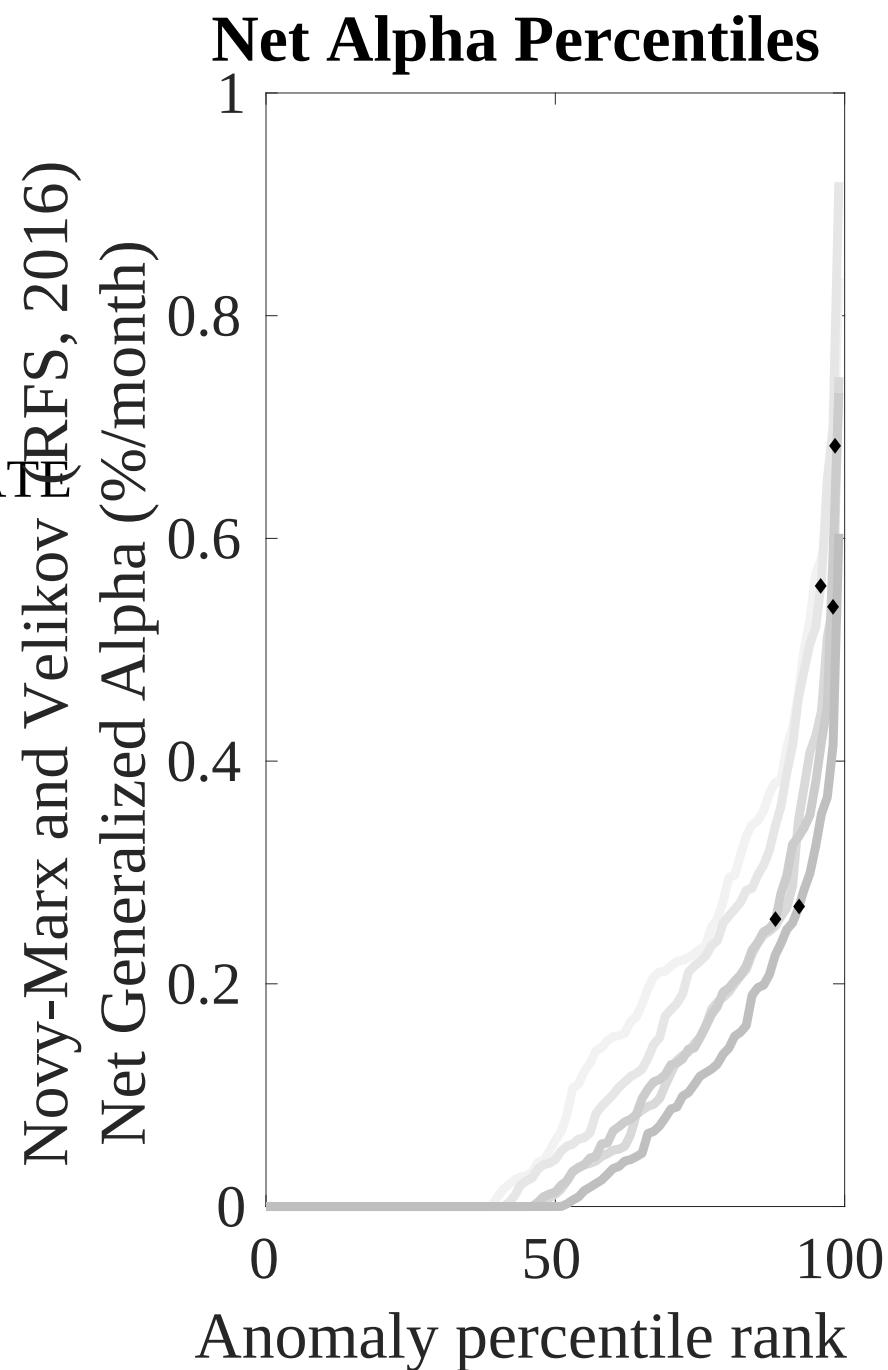
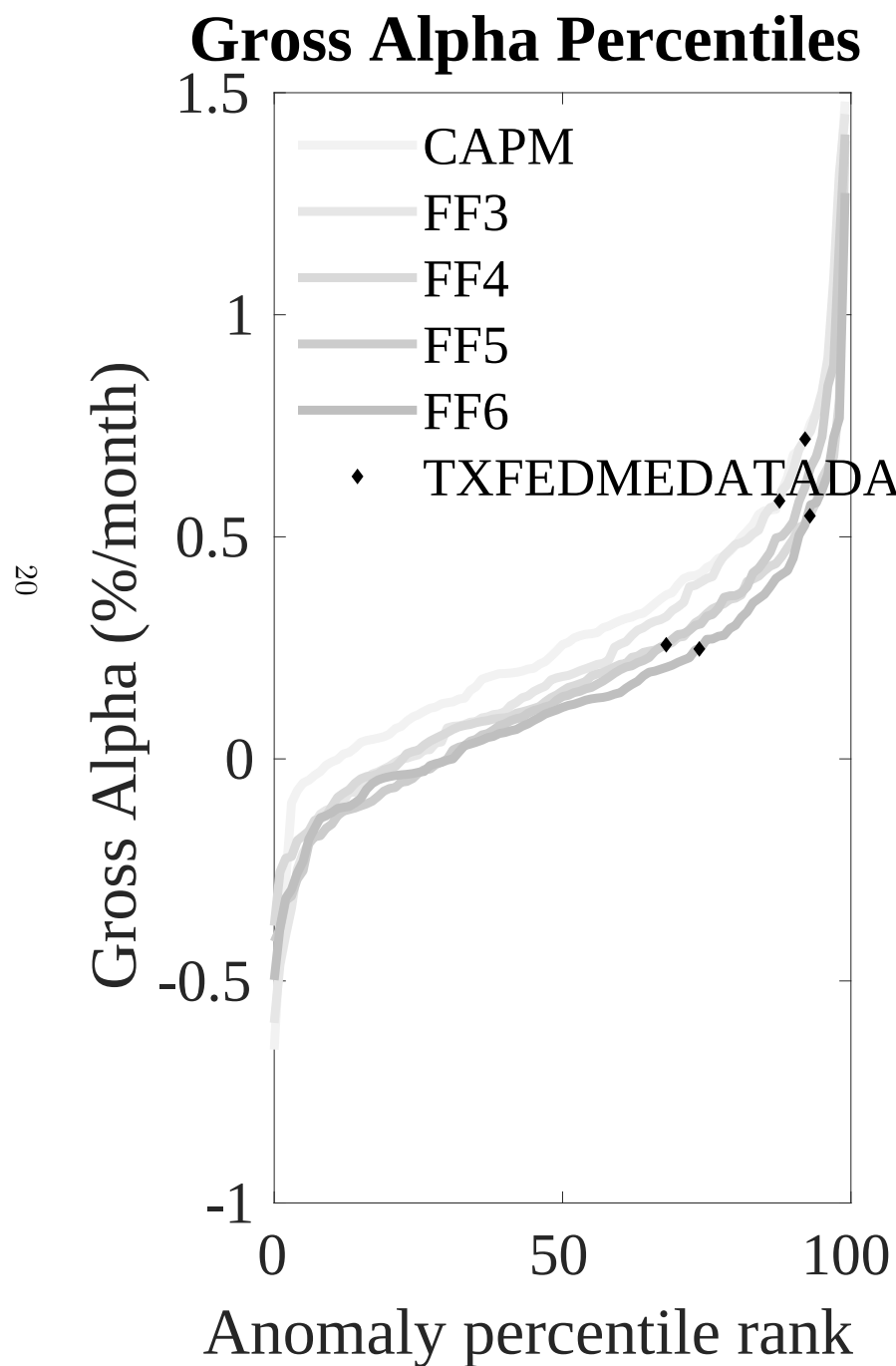


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TEB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



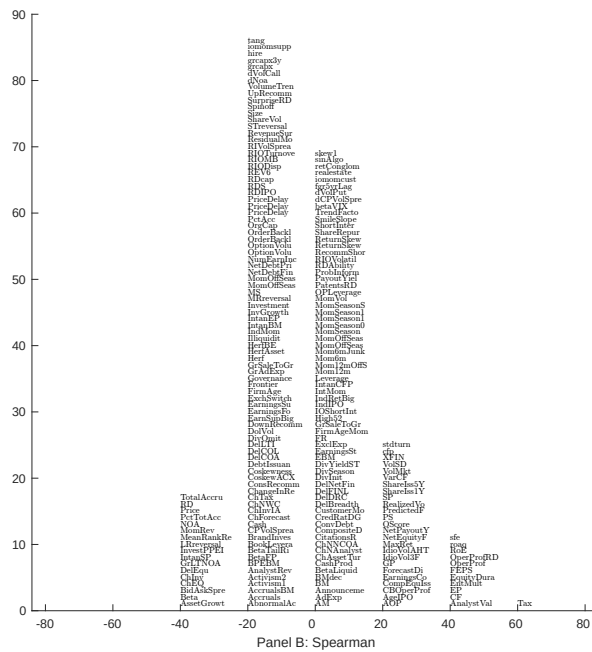
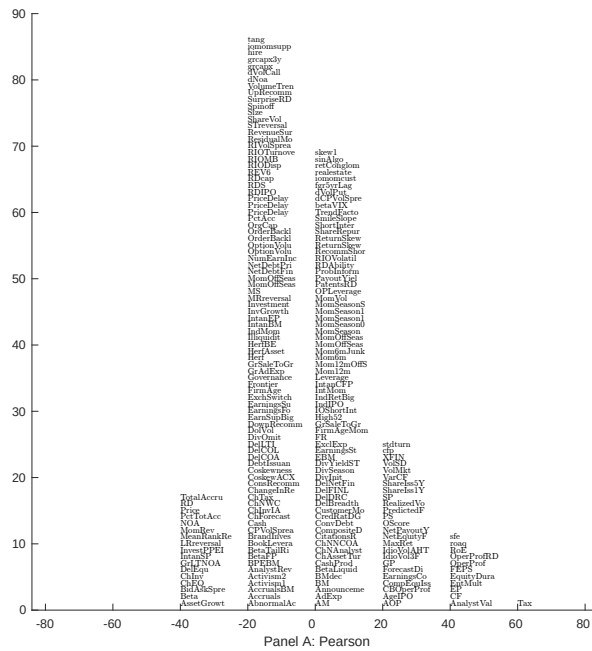


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with TEB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

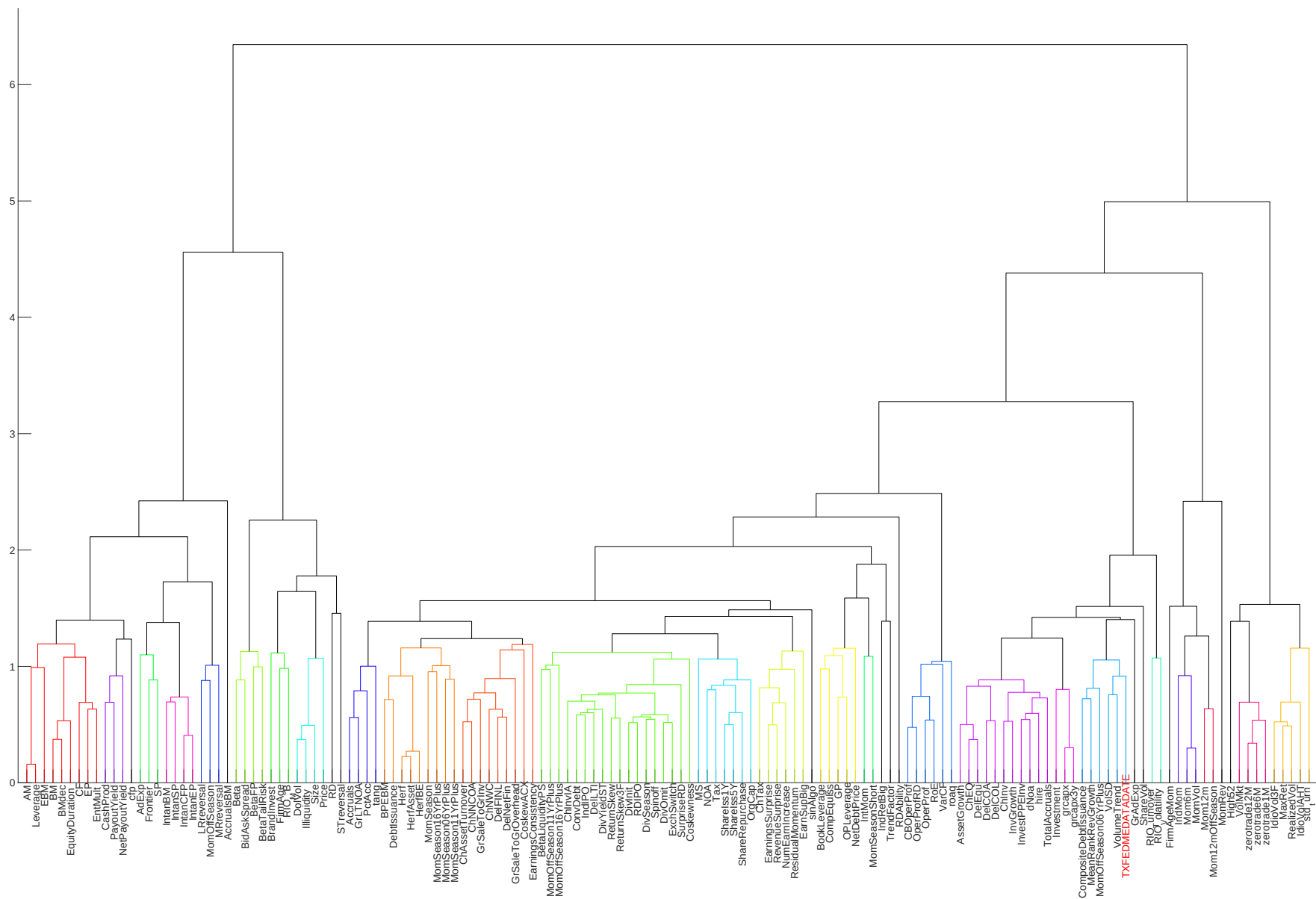


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

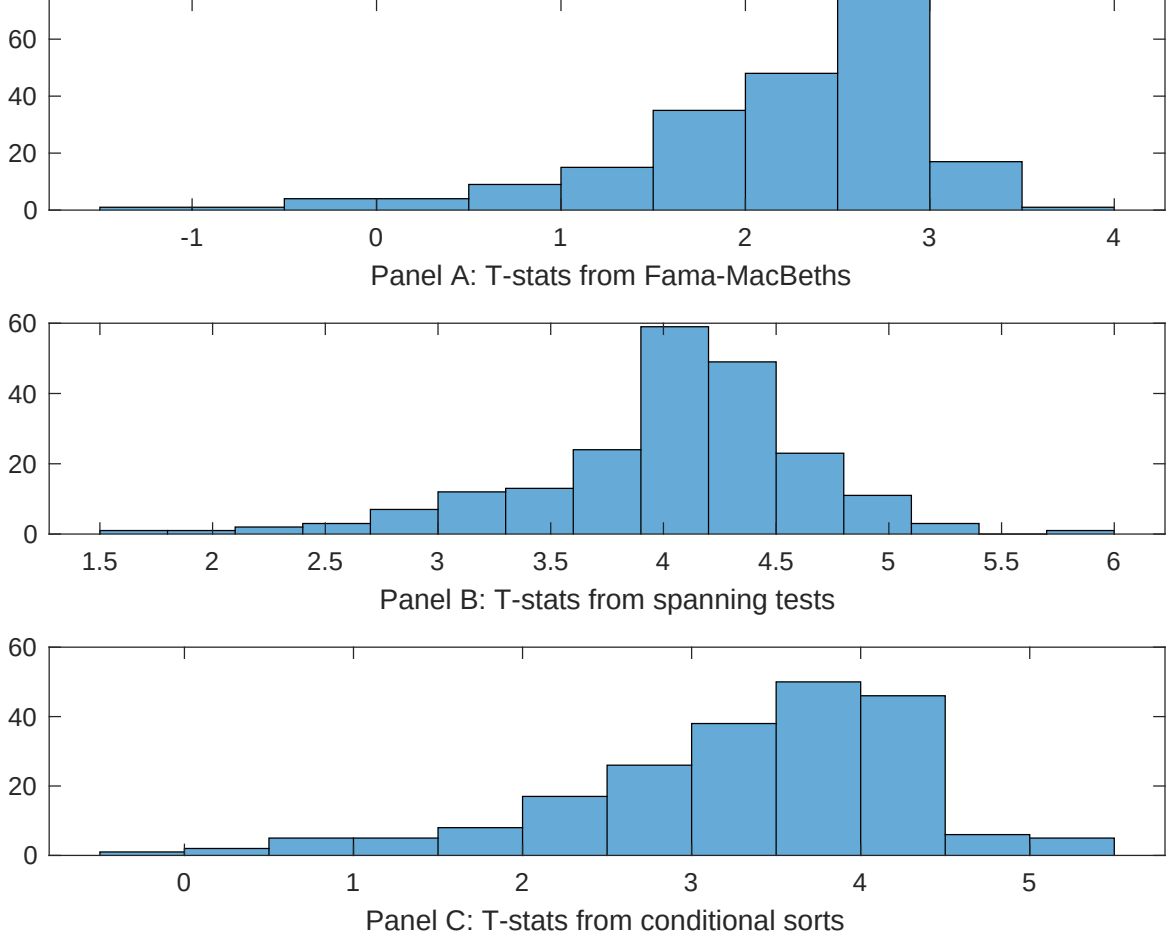


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TEB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TEB}TEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TEB. Stocks are finally grouped into five TEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TEB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TEB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TEB}TEB_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Analyst earnings per share, Net Payout Yield, EPS Forecast Dispersion, Realized (Total) Volatility, Net equity financing, Idiosyncratic risk (AHT). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.97 [3.26]	0.12 [4.85]	0.11 [4.36]	0.14 [7.87]	0.13 [5.04]	0.12 [6.65]	0.11 [5.91]
TEB	0.37 [2.52]	0.26 [1.53]	0.50 [2.50]	0.30 [2.65]	0.14 [0.98]	0.31 [3.16]	0.30 [1.30]
Anomaly 1	0.59 [1.09]						-0.20 [-0.08]
Anomaly 2		0.20 [2.11]					0.52 [1.14]
Anomaly 3			0.31 [2.05]				0.30 [3.09]
Anomaly 4				0.14 [3.44]			0.19 [5.51]
Anomaly 5					0.19 [3.25]		0.44 [1.09]
Anomaly 6						0.93 [1.56]	-0.18 [-2.48]
# months	576	668	576	679	629	679	576
$\bar{R}^2(\%)$	2	1	1	2	1	2	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TEB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TEB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Analyst earnings per share, Net Payout Yield, EPS Forecast Dispersion, Realized (Total) Volatility, Net equity financing, Idiosyncratic risk (AHT). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.28 [2.70]	0.30 [3.06]	0.25 [2.55]	0.25 [2.47]	0.26 [2.38]	0.24 [2.33]	0.26 [2.68]
Anomaly 1	15.71 [3.76]						-3.31 [-0.73]
Anomaly 2		35.65 [9.39]					29.43 [5.41]
Anomaly 3			29.48 [8.52]				27.32 [6.49]
Anomaly 4				19.42 [6.57]			7.23 [1.74]
Anomaly 5					16.36 [3.35]		-15.01 [-2.37]
Anomaly 6						16.68 [4.79]	-1.41 [-0.29]
mkt	-13.44 [-5.10]	-9.78 [-4.15]	-7.72 [-2.99]	-6.17 [-2.25]	-12.91 [-4.86]	-8.99 [-3.29]	-3.37 [-1.24]
smb	17.62 [3.75]	26.31 [7.70]	19.81 [5.08]	34.30 [8.43]	26.47 [6.26]	36.72 [7.48]	22.82 [4.73]
hml	19.21 [3.97]	14.34 [3.05]	26.92 [5.94]	22.50 [4.81]	27.53 [5.69]	24.05 [5.07]	15.99 [3.27]
rmw	69.41 [10.84]	65.67 [13.10]	59.86 [10.96]	71.25 [13.71]	78.96 [13.73]	71.56 [12.58]	51.26 [8.10]
cma	6.89 [0.97]	-17.89 [-2.46]	2.57 [0.38]	1.37 [0.20]	-1.16 [-0.15]	4.39 [0.62]	-12.88 [-1.76]
umd	1.44 [0.56]	3.65 [1.60]	-0.73 [-0.30]	-3.04 [-1.25]	1.43 [0.57]	-1.58 [-0.65]	0.13 [0.05]
# months	577	680	577	680	630	680	577
$\bar{R}^2(\%)$	55	54	59	51	50	50	62

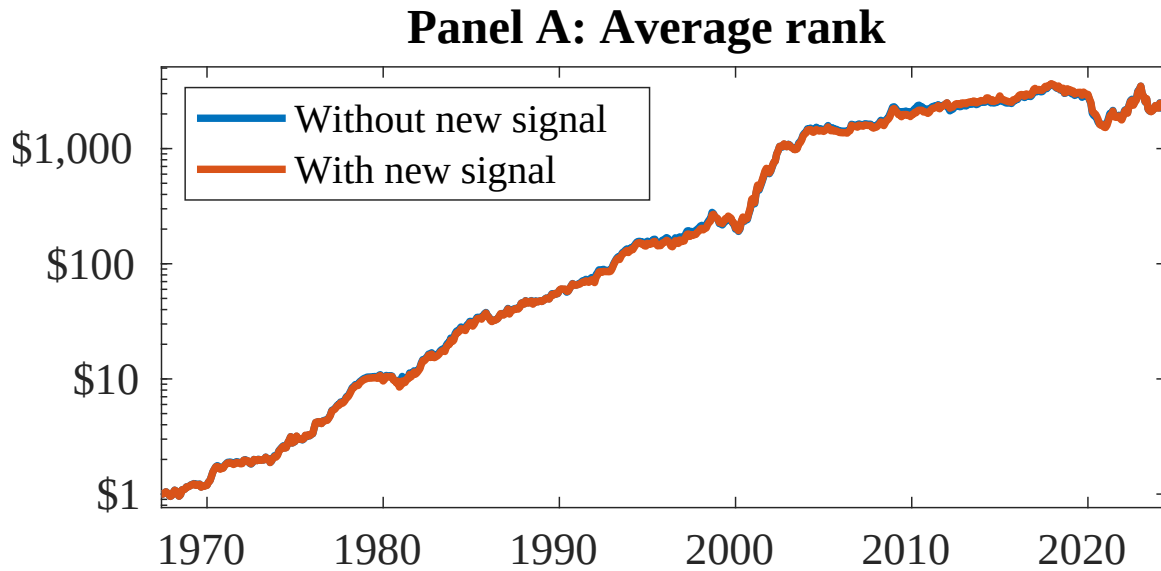


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as TEB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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